Experiment 10: Data Visualization for Inventory Management System

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Aim

To create data visualizations such as pie charts and bar graphs for an inventory management system to represent stock distribution effectively.

Objective

- To visualize inventory data using charts.
- To analyze stock distribution across categories.
- To improve data understanding through graphical representation.

Categories Used

- Electronics
- Clothing
- Home Appliances
- Books
- Toys

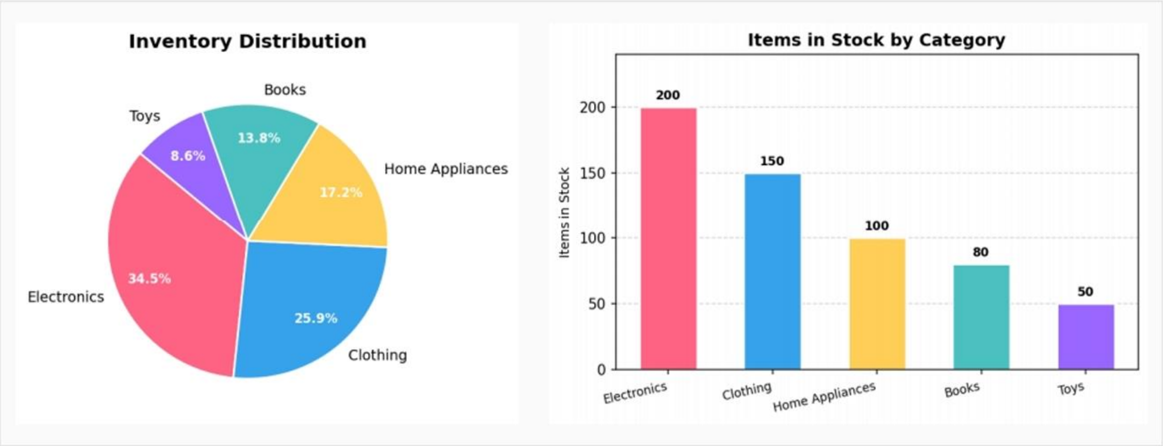
HCI Principles Used

- Visibility
- Simplicity
- Data clarity
- Consistency

Description

The experiment includes creating:

- Pie chart for inventory distribution
- Bar graph for category-wise stock analysis



Inventory Charts

The visualization helps users quickly understand stock availability in different product categories.

Python Code

```
import matplotlib.pyplot as plt

# Inventory data
categories = ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys']
stock = [200, 150, 120, 100, 80]

# Create figure
plt.figure(figsize=(10,5))

# Pie Chart
```

```
plt.subplot(1,2,1)
plt.pie(stock, labels=categories, autopct='%1.1f%%')
plt.title("Inventory Distribution")

# Bar Graph
plt.subplot(1,2,2)
plt.bar(categories, stock)
plt.title("Items in Stock by Category")
plt.xlabel("Categories")
plt.ylabel("Items in Stock")

# Display graphs
plt.tight_layout()
plt.show()
```

Working Process

1. Define inventory categories and stock values.
2. Create a pie chart to show stock distribution.
3. Create a bar graph for category comparison.
4. Display visualizations using Matplotlib.

Result

The pie chart and bar graph successfully represented inventory stock distribution and category-wise stock analysis.